Day 24: Overdispersion and the quasibinomial model

Last time we fit binomial logistic regression models and used the Pearson X^2 statistic to check whether the model fits. Today: what do we do when the goodness of fit test fails?

Recap: binomial logistic regression

```
binomial_glm <- glm(Y/n ~ predictors, weights = n,
                    data = dataset,
                    family = "binomial")
```

- **Binomial counts**: $Y_i \sim \text{Binom}(m_i, \pi_i)$
- **Independence**: observations are independent (Y_i's are independent *and* each 1/0 that contributes to Y_i is independent)
- **Linearity**: the log odds is a linear function of the predictors
- **Variance structure**: the variance of a binomial random variable is $n\pi(1-\pi)$

  $\implies$ so the variance of Y_i is $m_i\pi_i(1-\pi_i)$ (not constant!)

Data: early exposure to multiple languages

Fan et al. (2015) asked whether children who regularly *hear* other languages, even if they don't speak them, are better at taking another person's perspective. Children ages 4 to 6 played a game in which an adult asked them to move an object (for example, "I see a small car. Can you move the small car under the spoon?"). The adult could not see every object, so to succeed, the child had to think about which car the *adult* meant. Each child did 4 of these trials.

- `success`: number of the 4 trials the child got right
- `language`: `Monolingual`, `Exposed` (regularly hears another language but speaks only English), or `Bilingual`
- `age` (years), `sex`, `kbit` (a nonverbal IQ score), `ppvt` (a vocabulary score), and `maternal_ed` (maternal education)

Fan, S. P., Liberman, Z., Keysar, B., & Kinzler, K. D. (2015). The Exposure Advantage: Early Exposure to a Multilingual Environment Promotes Effective Communication. *Psychological Science*, 26(7), 1090–1097.

subject	age	sex	language	kbit	maternal_ed	ppvt	success	trials
1	5.0	0	Monolingual	107	1	128	4	4
3	4.4	1	Monolingual	112	1	104	3	4
5	4.0	0	Bilingual	112	1	111	1	4
6	5.1	0	Monolingual	110	0	105	0	4

subject	age	sex	language	kbit	maternal_ed	ppvt	success	trials
7	5.0	0	Bilingual	116	1	119	4	4
8	6.3	0	Monolingual	92	1	122	1	4

What is Y? What is n_i? What are the possible predictors? What is the primary predictor we want to make inference about?

term	estimate	std. error	z	p-value
(Intercept)	4.825	1.756	2.75	0.006
sex	0.185	0.282	0.65	0.513
kbit	0.010	0.010	0.93	0.352
ppvt	-0.018	0.011	-1.71	0.087
maternal_ed	0.142	0.351	0.40	0.686
age	-0.515	0.165	-3.12	0.002
languageExposed	-0.017	0.377	-0.04	0.965
languageMonolingual	-1.069	0.353	-3.03	0.002

Relative to _____ children, after accounting for all other variables in the model, _____ children are worse at the task.

Goodness of Fit Test

- H_0 :
- H_A :

```
X2 <- sum(residuals(lang_glm, type = "pearson")^2)
X2
```

```
[1] 143.4482
```

```
df.residual(lang_glm)
```

```
[1] 61
```

```
1 - pchisq(X2, df = 61)
```

```
[1] 1.358071e-08
```

Indicates the model is:

If the model is correct, the Pearson residuals should be approximately $N(0, 1)$:

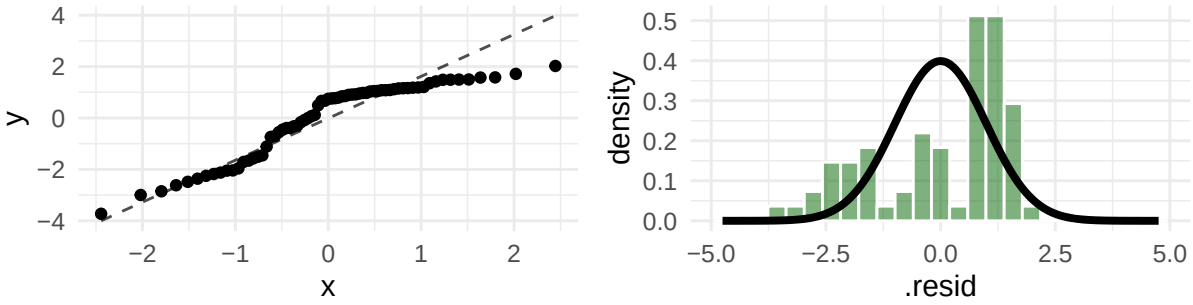
```
lang_aug <- augment(lang_glm, type.residuals = "pearson")

mean(lang_aug$.resid)
```

```
[1] -0.002170125
```

```
sd(lang_aug$.resid)
```

```
[1] 1.452421
```



These pearson residuals are _____

More on variance

In linear regression, the mean (μ_Y) and the variance (σ^2) are separate parameters. Points can be tightly clustered around the line or widely spread, and the model works either way.

In binomial logistic regression, the variance is determined by the mean:

$$E(Y_i) = n_i\pi_i \qquad Var(Y_i) = _____$$

> **Overdispersion**
>
> **Overdispersion** occurs when the observed variance in the data is _____ than the binomial model says it should be.

The binomial model assumes that each child's 4 trials are independent and have the same probability of success. Why might that not be true here? What would that do to the variance of Y_i?

Diagnosing overdispersion

If the model is correct, each Pearson residual is approximately $N(0, 1)$, so each squared Pearson residual is about _____ on average. That means $X^2 = \sum \text{Pres}_i^2$ should be close to the degrees of freedom.

> **Dispersion parameter**
>
> $$\widehat{\psi} = \frac{\sum \text{Pres}_i^2}{n - (p+1)} = \frac{X^2}{\text{residual df}}$$
>
> - $\widehat{\psi} \approx 1$: _____
> - $\widehat{\psi} \gg 1$: _____

For the language model: $\widehat{\psi} =$ _____ $\approx$ _____

Fixing overdispersion: the quasibinomial model

The **quasibinomial** model keeps the same model for π_i, but lets the variance be larger than the binomial variance:

$$Var(Y_i) = \psi \, n_i \pi_i (1 - \pi_i)$$

```
lang_qglm <- glm(success/trials ~ sex + kbit + ppvt + maternal_ed + age + language,
                 data = language,
                 family = "quasibinomial",
                 weights = trials)

summary(lang_qglm)$dispersion
```

```
[1] 2.351658
```

This is the same $\widehat{\psi}$ we calculated from the Pearson residuals.

Comparison of binomial and quasibinomial estimates:

term	estimate (binomial)	SE (binomial)	estimate (quasi)	SE (quasi)
(Intercept)	4.825	1.756	4.825	2.693
sex	0.185	0.282	0.185	0.433
kbit	0.010	0.010	0.010	0.016
ppvt	-0.018	0.011	-0.018	0.017
maternal_ed	0.142	0.351	0.142	0.539
age	-0.515	0.165	-0.515	0.253
languageExposed	-0.017	0.377	-0.017	0.578
languageMonolingual	-1.069	0.353	-1.069	0.541

What changed when we switched to the quasibinomial model? What stayed the same?

Inference with the quasibinomial model

Because we *estimate* ψ, we go back to t and F distributions (like in linear regression, where we estimated σ).

i One coefficient: t-test

Test statistic: $t = \dfrac{\hat{\beta} - 0}{SE_{\text{quasi}}(\hat{\beta})} = \dfrac{\hat{\beta} - 0}{\sqrt{\widehat{\psi}}\, SE_{\text{binom}}(\hat{\beta})}$

Reference distribution: t with $df =$ _____

i Several coefficients: drop-in-deviance F-test

Hypotheses: $H_0 : \beta_i = \beta_j = \cdots = 0$ vs. H_a: at least one $\neq 0$

Test statistic: $F = \dfrac{(\text{Deviance}_{\text{reduced}} - \text{Deviance}_{\text{full}})/d}{\widehat{\psi}}$, where $d =$ _____

Reference distribution: F with d and $n - (p+1)$ df (residual df of the full model)

In R, use `anova(reduced, full, test = "F")`.

Is `language` needed in the model?

- H_0 :
- H_A :

```r
reduced_qglm <- glm(success/trials ~ sex + kbit + ppvt + maternal_ed + age,
                    data = language,
                    family = "quasibinomial",
                    weights = trials)

anova(reduced_qglm, lang_qglm, test = "F")
```

```
Analysis of Deviance Table

Model 1: success/trials ~ sex + kbit + ppvt + maternal_ed + age
Model 2: success/trials ~ sex + kbit + ppvt + maternal_ed + age + language
  Resid. Df Resid. Dev Df Deviance      F  Pr(>F)
1        63     180.58
2        61     167.35  2   13.237 2.8145 0.06774 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

What about the effect of monolinguism?

```r
exp(confint(lang_qglm))
```

```
                        2.5 %       97.5 %
(Intercept)         0.7790849 3.523281e+04
sex                 0.5135667 2.828747e+00
kbit                0.9785628 1.041248e+00
ppvt                0.9490896 1.013455e+00
maternal_ed         0.3930132 3.315421e+00
age                 0.3576912 9.715270e-01
languageExposed     0.3130941 3.078133e+00
languageMonolingual 0.1148159 9.750352e-01
```

We are 95% confident that the _____ for _____ children is somewhere between ____% to ____% smaller than for _____, holding the other variables constant.

With the binomial model, this interval was (0.17, 0.68). Why is the quasibinomial interval wider? Which one should we report?

Strategy for binomial logistic regression

1. Make sure the data are grouped appropriately (each row is independent).
2. Start by fitting `family = "quasibinomial"`, and look at $\widehat{\psi}$:
   - If $\widehat{\psi} > 1$: _____
   - If $\widehat{\psi} \approx 1$: _____
   - If $\widehat{\psi} < 1$: _____

Summary

- **Overdispersion**: the data are more variable than the binomial model allows, so the binomial standard errors are too _____.
- Estimate the dispersion with $\widehat{\psi} = X^2/(n - (p + 1))$, the sum of the squared Pearson residuals divided by the residual df.
- The **quasibinomial** model gives the same estimates, but multiplies the standard errors by _____.
- With a quasibinomial model, use _____ tests for one coefficient and drop-in-deviance _____ tests for several.

R function reference

Function	Purpose
`glm(y/n ~ x, weights = n, data, family = "binomial")`	Fit a binomial logistic regression
`sum(residuals(model, type = "pearson")^2)`	Pearson X^2 statistic
`glm(y/n ~ x, weights = n, data, family = "quasibinomial")`	Fit a quasibinomial model
`summary(model)$dispersion`	Estimated dispersion parameter $\widehat{\psi}$
`anova(reduced, full, test = "F")`	Drop-in-deviance F-test for a quasibinomial model
`exp(confint(model))`	Confidence intervals on the odds ratio scale